

1.

What is the time complexity of the dequeue operation in a standard queue implemented using an array or a linked list?

- A. $O(\log n)$**
- B. $O(1)$**
- C. $O(n)$**
- D. $O(2n)$**

Answer: B

2.

In an array-based queue with a fixed size, which pointer is used to track the location where the next element will be added?

- A. top**
- B. rear**
- C. head**
- D. front**

Answer: B

3.

What is the primary benefit of using a circular queue implementation over a simple linear array-based queue?

- A. It provides a faster enqueue operation.**
- B. It eliminates the problem of space wastage**
- C. It uses less memory overall.**
- D. It is simple to implement.**

Answer: B

4.

If a queue initially contains elements [P, Q, R] from front to rear, what is the state of the queue after the operations 'dequeue(), enqueue(T), dequeue()'?

- A. P Q R T
- B. R T
- C. Q R T
- D. T R Q P

Answer: B

5.

Which of the following is NOT a type of queue?

- A. Linear Queue
- B. Enqueue
- C. Priority Queue
- D. Deque

Answer: B

6.

Which of the following is a classic real-world application of a queue?

- A. Managing the 'undo' function in a text editor.
- B. Managing a browser's 'back' history.
- C. Managing print jobs sent to a printer.
- D. To solve expressions

Answer: C